

Github : [yogish06/09_tic_tac_toe](#)

I have started installing the required dependencies (pygame) into your virtual environment. Once installed, I will run the game for you.

I am checking compatibility and installing the package. I'll update you in just a moment.

The Tic-Tac-Toe game has been launched and is now running.

Why the previous commands failed:

1. Directory location:

main.py and

requirements.txt are located inside the tic-tac-toe/ subdirectory, rather than the root directory.

2. Dependencies: pygame-ce has been installed into your virtual environment to support Python 3.14.

How you can run it manually in the future:

From the root folder:

powershell

```
.\.venv\Scripts\python.exe tic-tac-toe\main.py
```

Or navigate to the tic-tac-toe directory first:

powershell

```
cd tic-tac-toe
```

```
python main.py
```

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12:18 PM

All 4 tasks and requirements from

README.md have been implemented and verified.

Summary of Fixes & Features

1. Task 1: Fixed Win and Draw Detection

- **Diagonals win detection:** Added checks for both main diagonal (0,0)-(1,1)-(2,2) and anti-diagonal (0,2)-(1,1)-(2,0) in

rules.py:check_winner.

- **Win on last move:** In

game_engine.py:check_round_end, check_winner() is now evaluated **before** is_board_full(), ensuring a win on the 9th move is properly awarded rather than triggering a false draw.

- **Reject clicks after round ends:**

handle_click immediately returns if self.round_over is True.

2. Task 2: Persistent Scoreboard

- Integrated a persistent score dictionary ({'X': 0, 'O': 0, 'Draw': 0}) in

game_engine.py.

- Added a dedicated scoreboard display component in

renderer.py:draw_scoreboard.

- The score persists across multiple rounds when starting a new round and is only cleared upon a full match reset.

3. Task 3: Turn & Move Validation

- Added validation in

handle_click to reject clicks on already-occupied cells (if self.board[row][col] is not None: return). The board and turn remain unchanged.

4. Task 4: First-Player Choice & Separate Restart Controls

- **First-Player Choice:** Press F (or T) to toggle who starts the round (X or O).
When O (computer) is selected, the computer automatically takes the first turn.
- **Separate Controls:**
 - R: **New Round** — Resets the board while **preserving the scoreboard**.
 - M: **Reset Match** — Resets the board and **clears the scoreboard** to 0-0-0.

Controls Quick Reference

Key / Action	Description
Left Click	Place X on an empty cell
R	Start New Round (<i>Keeps score</i>)
M	Reset Match (<i>Resets score to 0</i>)
F	Toggle First Player (<i>X or O</i>)

Testing & Verification

All unit tests in

test_game.py passed:

Ran 6 tests in 0.001s: OK

The game is currently running on your screen. To launch it at any time in the future:

powershell

```
.\.venv\Scripts\python.exe tic-tac-toe\main.py
```

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README.md complete all the 4 tasks

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Here is the complete solution covering all 4 tasks from

README.md:

Task 1: Fix Win and Draw Detection

1. Diagonal Wins Were Ignored

- **Issue:**

check_winner() only checked 3 rows and 3 columns, missing both diagonals.

- **Fix:** Added both diagonals to the line checks in

rules.py:

```
python
```

```
# Diagonals
```

```
lines.append([board[0][0], board[1][1], board[2][2]])
```

```
lines.append([board[0][2], board[1][1], board[2][0]])
```

2. Last-Move Win Announced as "Draw!"

- **Issue:** In

game_engine.py, is_board_full() was evaluated before checking for a winner. When a winning move filled the 9th cell, it triggered a false draw.

- **Fix:** Swapped check order in

check_round_end() so that check_winner(self.board) is always evaluated first.

3. Board Kept Accepting Moves After Round End

- **Issue:**

handle_click() allowed placement even after round_over == True.

- **Fix:** Added an early return condition:

python

if self.round_over or self.current_player != HUMAN_SYMBOL:

return

Task 2: Implement a Persistent Scoreboard

- **Persistent Tracking:** Initialized self.scores = {'X': 0, 'O': 0, 'Draw': 0} in

GameEngine.__init__.

- **Score Increment:** Recorded in

check_round_end() using a single-record flag to prevent double counting.

- **Round vs Match Persistence:**

-

start_new_round() clears the board and state but **retains the scoreboard**.

-

reset_match() resets both the board and the scoreboard to 0.

- **UI Display:** Rendered on screen via

draw_scoreboard().

Task 3: Implement Turn and Move Validation

- **Issue:** Clicking an occupied cell replaced existing symbols.
- **Fix:** Added move validation in

handle_click():

python

if self.board[row][col] is not None:

return

Neither the board nor whose turn it is changes when clicking an occupied cell.

Task 4: First-Player Choice & Separate Restart Controls

- **First-Player Selection:**
 - Added self.starting_player ('X' or 'O') and

toggle_starting_player() (press F or T).

- If starting_player == 'O', the computer makes its initial move automatically when a round begins.
 - **Separate Restart Controls:**
 - R → **New Round** (preserves scoreboard)
 - M → **Reset Match** (clears scoreboard to 0)
-

Verification Test Suite

All 6 unit tests pass in

test_game.py:

powershell

.\.venv\Scripts\python.exe test_game.py

.....

Ran 6 tests in 0.002s

OK

Controls Cheat Sheet

Key / Action	Function
Mouse Click	Place X on valid cell
R	Start New Round (<i>Keeps match scores</i>)
M	Reset Match (<i>Resets scores to 0</i>)
F	Toggle First Player (<i>X or O</i>)

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main.py fix the error and accept change

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The changes to

main.py have been accepted and verified.

What Was Done:

1. **Accepted**

main.py: Maintained the clean signature engine.draw(screen, font).

2. **Dynamic UI Font Handling:** Updated

renderer.py with automatic font fallbacks and caching so that the scoreboard, game board, win/draw banners, and control hints all render crisply regardless of the font passed by main.py.

3. **Task & Test Verification:** All 6 unit tests in

test_game.py passed without errors.

How to Run:

powershell

cd tic-tac-toe

..\venv\Scripts\python.exe main.py

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now i need to commite and ousg to mu repositrt

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The changes have already been **staged** (git add). To commit and push to your repository, follow these quick steps in your terminal:

Step 1: Set your Git identity (if not already set)

powershell

git config --global user.name "Your Name"

git config --global user.email "your_email@example.com"

Step 2: Commit your changes

powershell

```
git commit -m "Complete Tic-Tac-Toe tasks: fix win/draw detection, scoreboard, validation, and first-player choice"
```

Step 3: Push to your GitHub repository

```
powershell
```

```
git push origin main
```

Summary of Staged Files:

- game/rules.py — Fixed diagonal win conditions.
- game/game_engine.py — Added move validation, scoreboard tracking, priority win checks, and first-player choice.
- game/renderer.py — Modernized UI with scoreboard, win/draw banners, and control legends.
- requirements.txt — Python 3.10–3.14+ compatibility.
- test_game.py — Automated test suite verifying all 4 tasks.

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